CHAPTER 4:

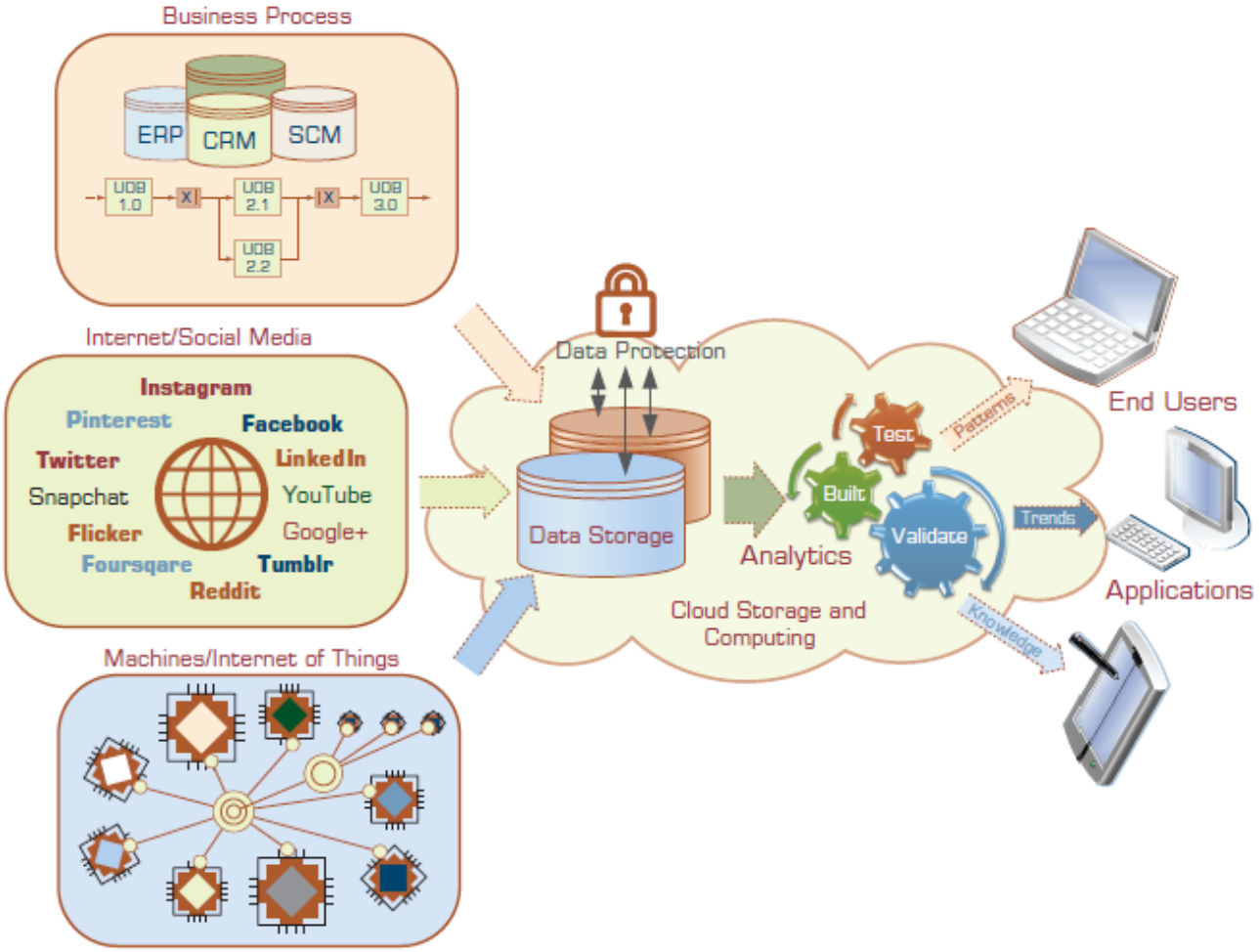
DATA ANALYTIC APPLICATION FOR BUSINESS

(Part 1): ETL and Data Pre-processing

The Nature of Data

- Data: a collection of facts
  - ✓ usually obtained as the result of experiences, observations, or experiments.
- Data may consist of text, numbers, figures, audio, video, images, etc.
- Data is the lowest level of abstraction (from which information and knowledge are derived).
- Data is the source for information and knowledge.
- The process from data to knowledge has to be handle carefully:
  - ✓ To ensure of its quality (garbage-in-garbage-out, GIGO)
  - ✓ Includes: identifying the data source, collecting, pre-process and modelling the data before presenting the outcomes within a context.

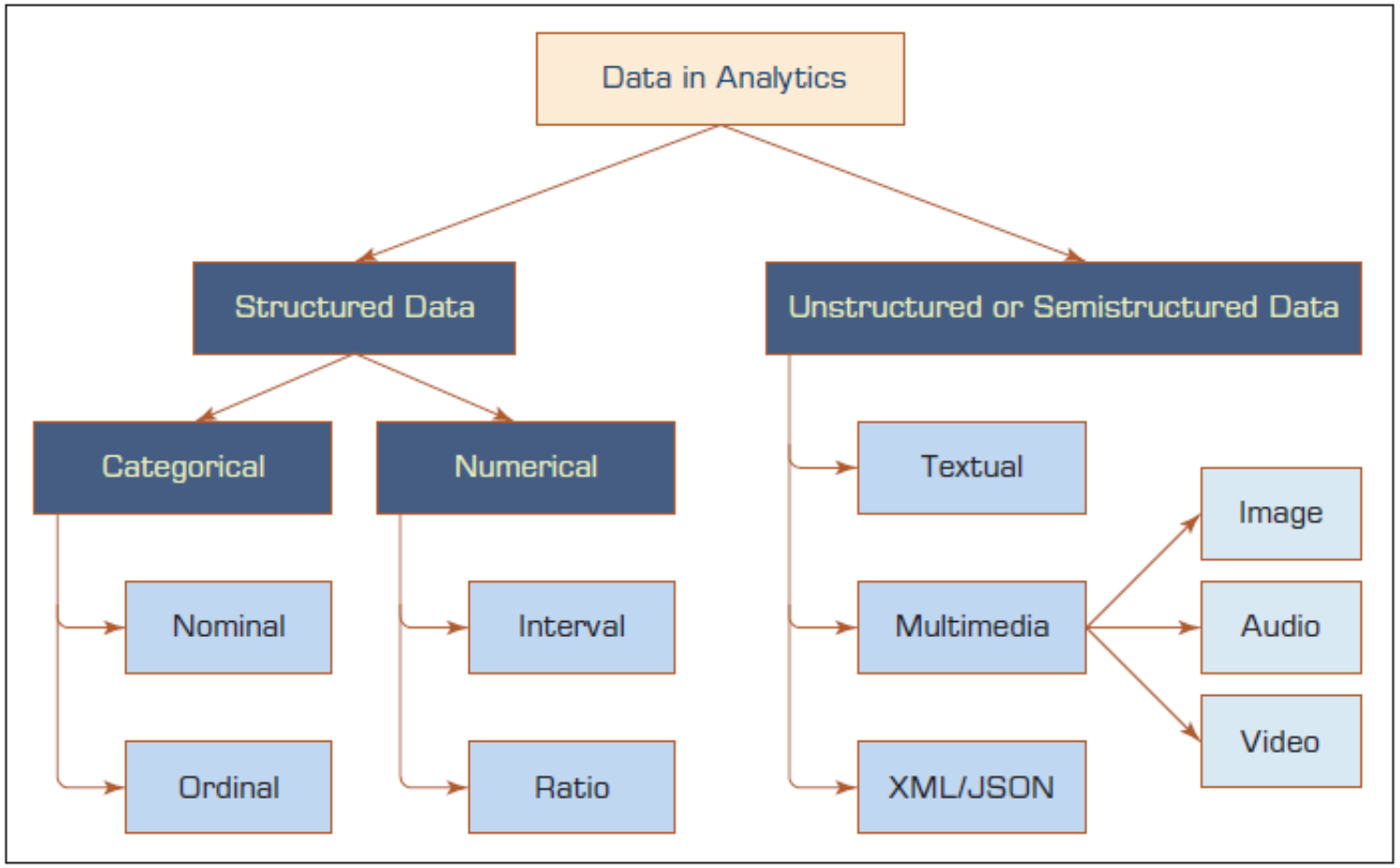
The Nature of Data



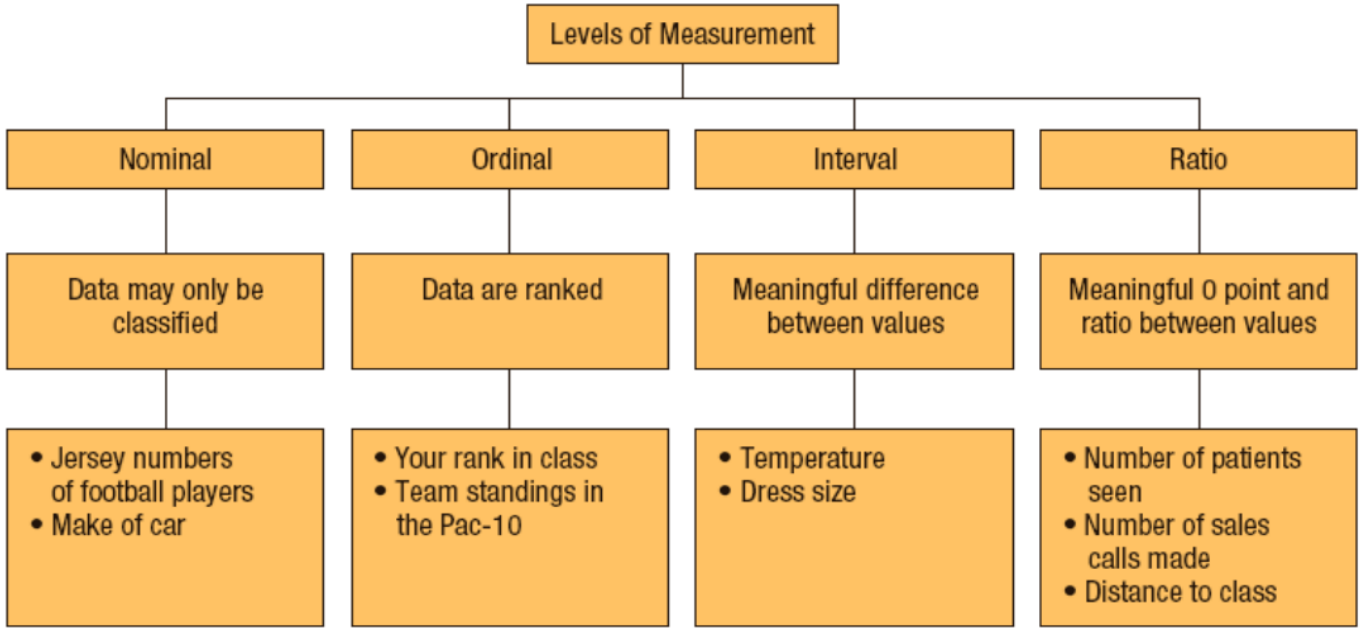
A Simple Taxonomy of Data

- Data (datum—singular form of data): facts
- Structured data
  - ✓Targeted for computers to process
  - ✓Numeric versus nominal
- Unstructured/textual data
  - ✓Targeted for humans to process/digest
- Semi-structured data?
  - ✓XML, HTML, Log files, etc.

A Simple Taxonomy of Data



Data: Level of Measurement



Metrics for Analytics Ready Data

- Data source reliability
- Data content accuracy
- Data accessibility
- Data security and data privacy
- Data richness
- Data consistency
- Data currency/data timeliness
- Data granularity
- Data validity and data relevancy

Data Pre-processing

- The real-world data is dirty, misaligned, overly complex, and inaccurate:
  - ✓ incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
  - ✓ noisy: containing errors or outliers
  - ✓ inconsistent: containing discrepancies in codes or names

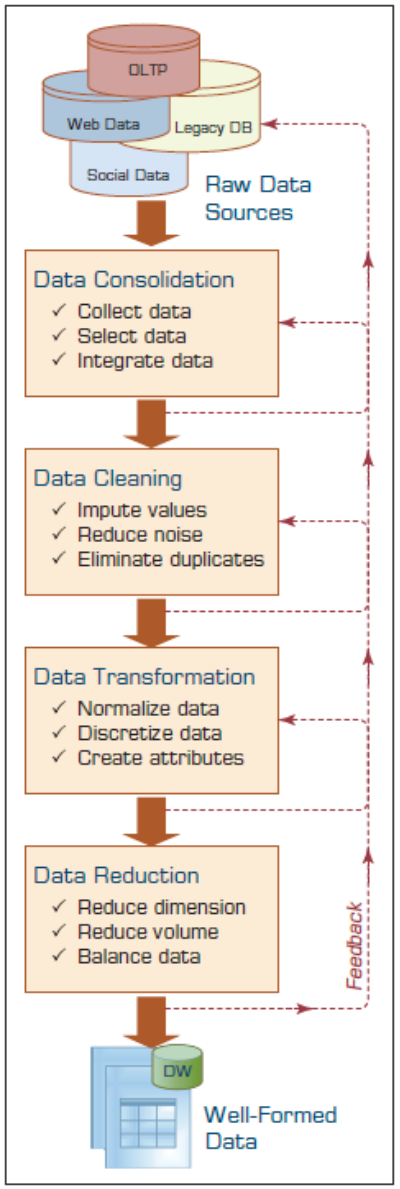
  => Not ready for analytics!
- No quality data, no quality mining results!
  - ✓ Quality decisions must be based on quality data
  - ✓ Data warehouse needs consistent integration of quality data
- Readying the data for analytics is needed.

Data Pre-processing

- Process involve to prepare the data for analytics:
  - ✓ Data preprocessing
  - ✓ Data consolidation
  - ✓ Data cleaning
  - ✓ Data transformation
  - ✓ Data reduction

Improving Analytics Readiness

These 4 steps are termed:
Data Pre-processing Steps →



Data reduction:

1. Variables
    - Dimensional reduction
    - Variable selection
2. Cases/samples
    - Sampling
    - Balancing / stratification

Data cleaning:

1. Fill in missing values
2. Identify outliers and smooth out noisy data
3. Correct inconsistent data

Missing Data

- Data is not always available
  - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data.
- Missing data may be due to:
  - ✓ equipment malfunction
  - ✓ inconsistent with other recorded data and thus deleted
  - ✓ data not entered due to misunderstanding
  - ✓ certain data may not be considered important at the time of entry
  - ✓ not register history or changes of the data
- Missing data may need to be inferred.

Missing Data

- In federated data, between 30%-70% of the data points will have at least one missing attribute - data wastage if we ignore all records with a missing value
- Remaining data is seriously biased.
- Lack of confidence in results.
- Understanding pattern of missing data unearths data integrity issues.

How to Handle Missing Data?

Among the options:

a) Ignore the tuple: Usually done when class label is missing (assuming the tasks in classification—not effective when the percentage of missing values per attribute varies considerably.

b) Fill in the missing value manually: tedious + infeasible?

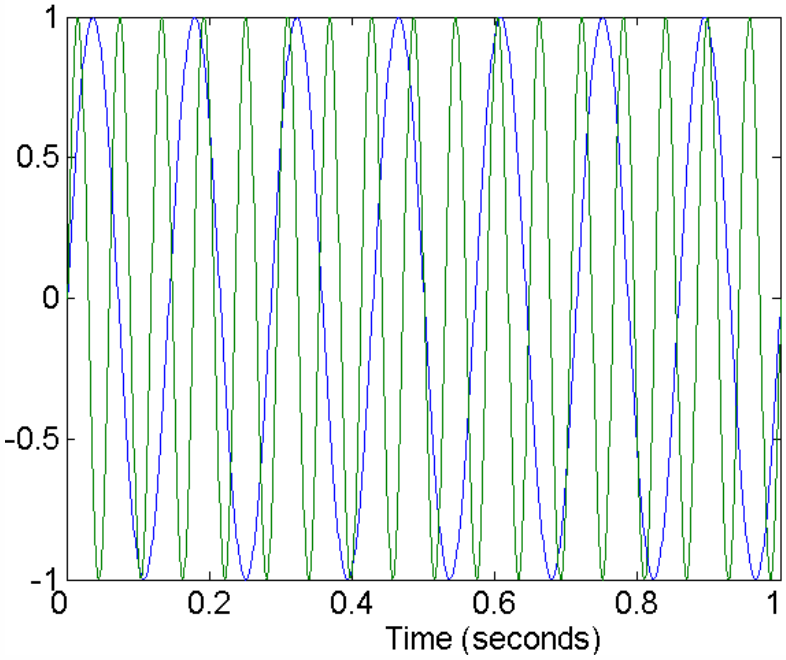
c) Use a global constant to fill in the missing value: e.g., “unknown”, a new class?

d) Fill in missing values (imputations) with most appropriate values (mean, median, min/max, mode, etc.)

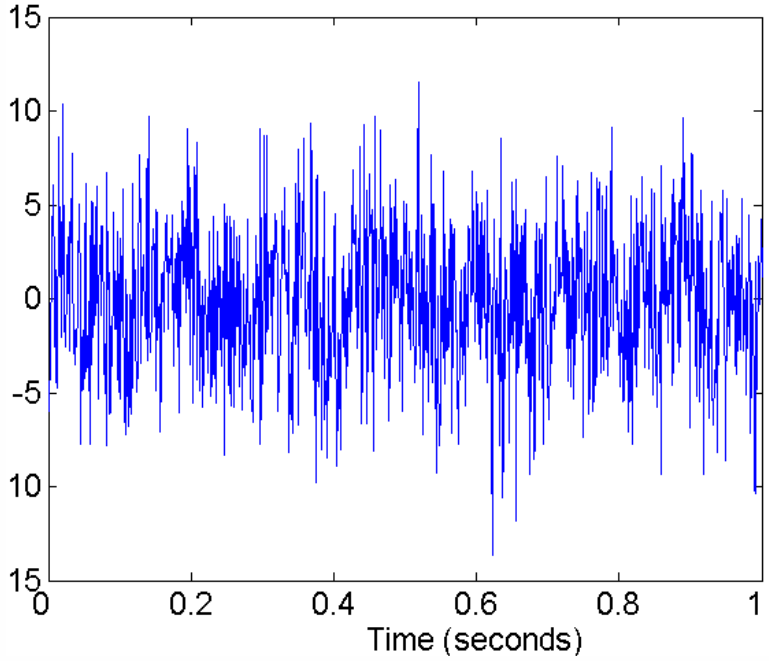
e) Remove the record of the missing value.

Noise

- Noise refers to modification of original values
  - Examples: distortion of a person's voice when talking on a poor phone



Two Sine Waves



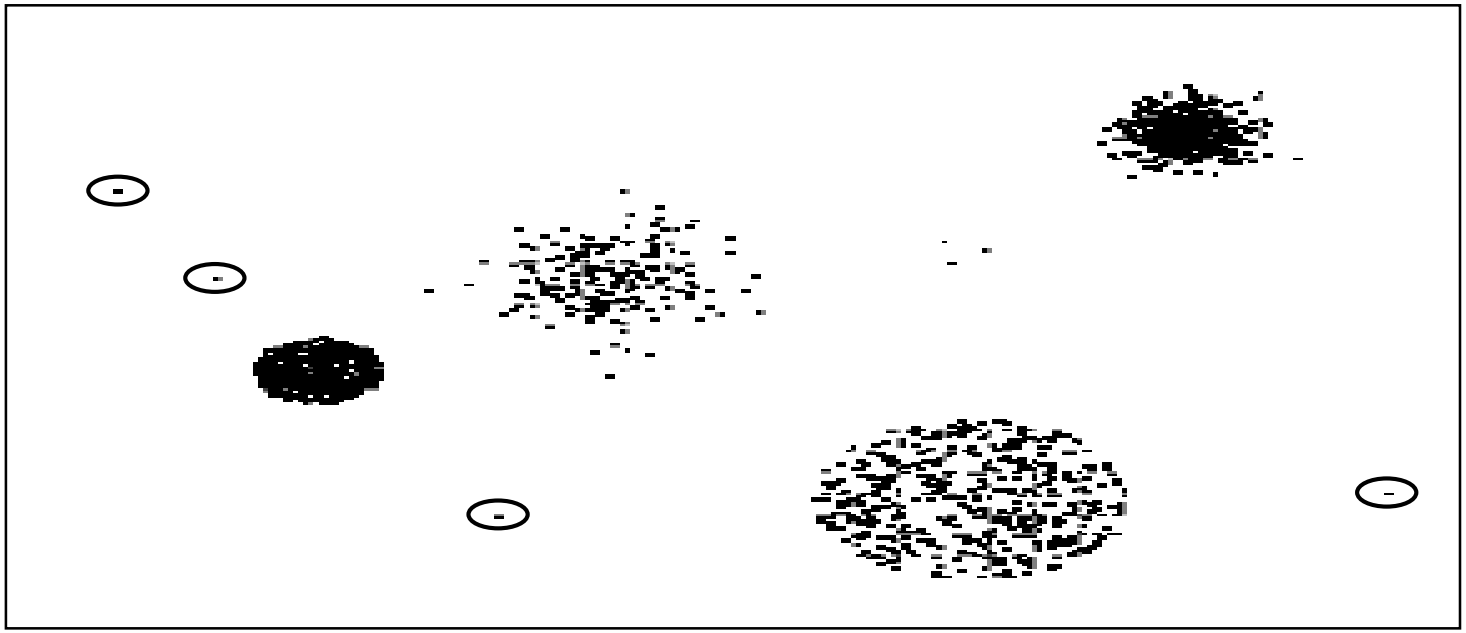
Two Sine Waves + Noise

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may due to:
  - ✓ faulty data collection instruments
  - ✓ data entry problems
  - ✓ data transmission problems
  - ✓ technology limitation
  - ✓ inconsistency in naming convention

Outliers

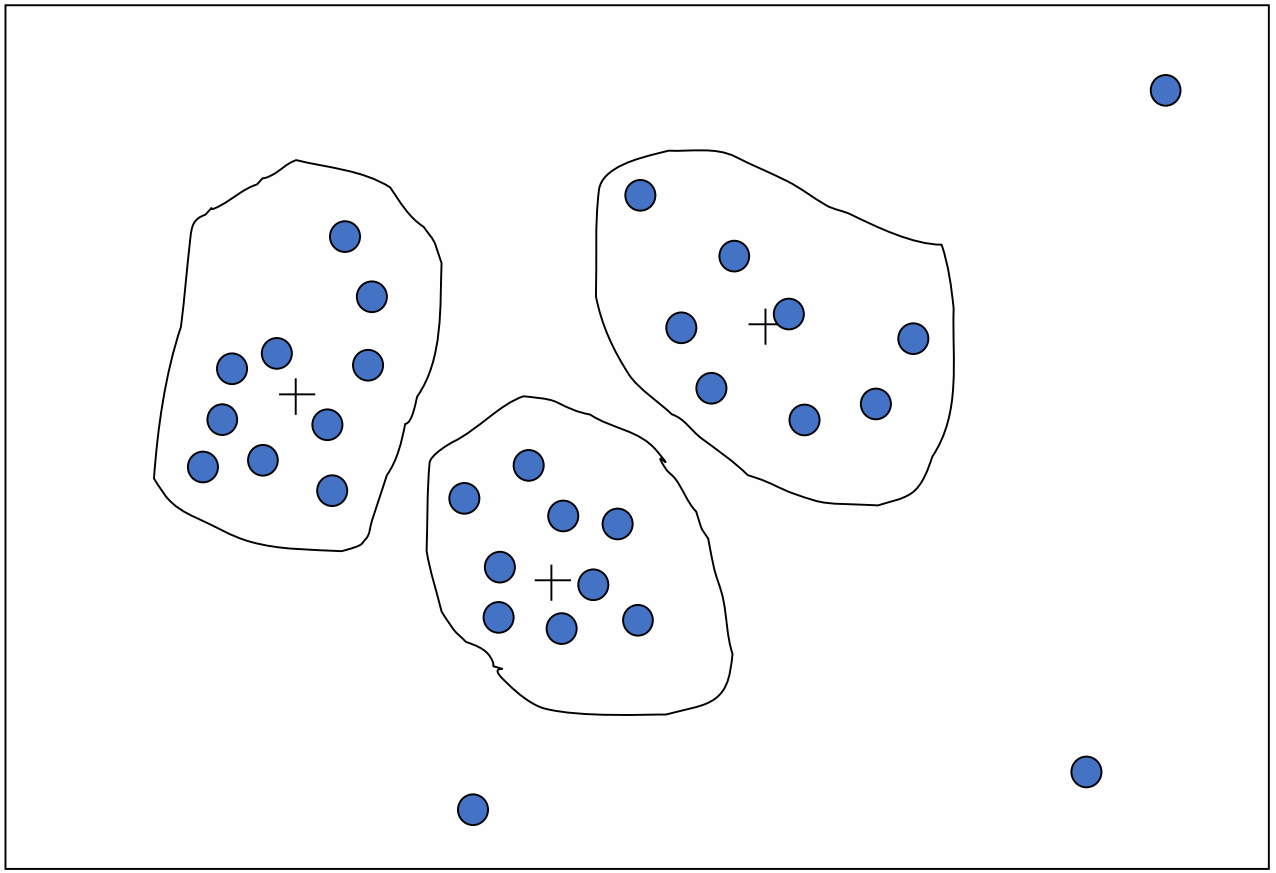
- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set.



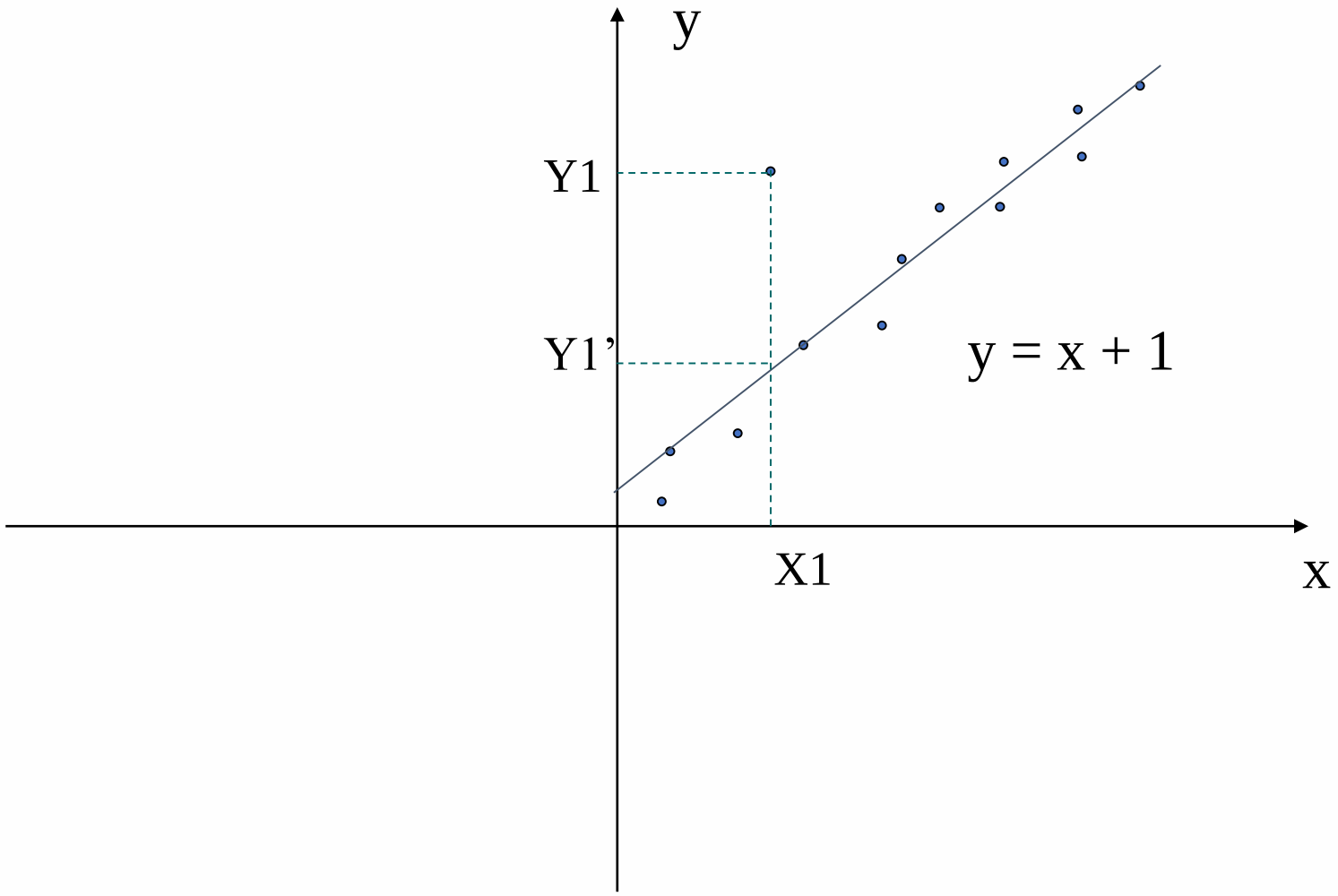
How to Handle Noisy Data?

- Binning method:
  a) first sort data and partition into (equal-depth) bins
  b) then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Clustering:
  ✓ detect and remove outliers.
- Combined computer and human inspection:
  ✓ detect suspicious values and check by human.
- Regression:
  ✓ smooth by fitting the data into regression functions.

Cluster Analysis



Regression



Data Preprocessing Tasks and Methods

Table 1: A summary of data pre-processing tasks and potential methods

Main Task	Subtasks	Popular Methods
Data consolidation	Access and collect the data Select and filter the data Integrate and unify the data	SQL queries, software agents, Web services. Domain expertise, SQL queries, statistical tests. SQL queries, domain expertise, ontology-driven data mapping.
Data cleaning	Handle missing values in the data	Fill in missing values (imputations) with most appropriate values (mean, median, min/max, mode, etc.); recode the missing values with a constant such as “ML”; remove the record of the missing value; do nothing.
	Identify and reduce noise in the data	Identify the outliers in data with simple statistical techniques (such as averages and standard deviations) or with cluster analysis; once identified, either remove the outliers or smooth them by using binning, regression, or simple averages.

Main Task	Subtasks	Popular Methods
Data cleaning	Find and eliminate erroneous data	Identify the erroneous values in data (other than outliers), such as odd values, inconsistent class labels, odd distributions; once identified, use domain expertise to correct the values or remove the records holding the erroneous values.
Data transformation	Normalize the data	Reduce the range of values in each numerically valued variable to a standard range (e.g., 0 to 1 or -1 to +1) by using a variety of normalization or scaling techniques.
	Discretize or aggregate the data	If needed, convert the numeric variables into discrete representations using range-or frequency-based binning techniques; for categorical variables, reduce the number of values by applying proper concept hierarchies.

Main Task	Subtasks	Popular Methods
Data transformation	Construct new attributes	Derive new and more informative variables from the existing ones using a wide range of mathematical functions (as simple as addition and multiplication or as complex as a hybrid combination of log transformations).
Data reduction	Reduce number of attributes	Principal component analysis, independent component analysis, chi-square testing, correlation analysis, and decision tree induction.
	Reduce number of records	Random sampling, stratified sampling, expert-knowledge-driven purposeful sampling.
	Balance skewed data	Oversample the less represented or undersample the more represented classes.